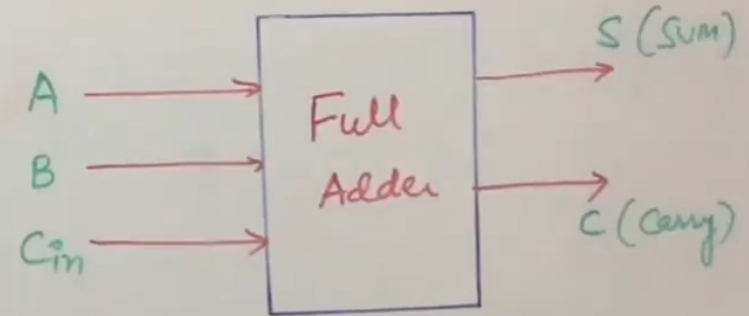


$$2^n = 2^3 = 8$$

Designing of a Full Adder

A	B	C _{in}	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$2^n = 2^3 = 8$$

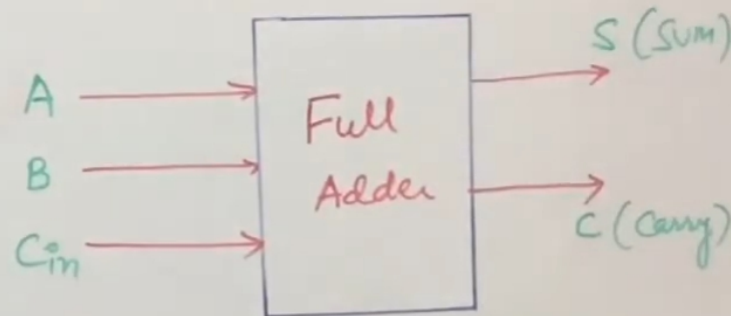
Designing of a Full Adder

A	B	C _{in}	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-Map for S

		AB			
	C_{in}	00	01	11	10
0		0 ₀	1 ₂	0 ₄	1 ₅
1		1 ₁	0 ₃	1 ₇	0 ₆

$$\begin{aligned}
 S &= \bar{A}\bar{B}\bar{C}_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + A\bar{B}C_{in} \\
 &= \bar{A}\bar{B}C_{in} + AB\bar{C}_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}C_{in} \\
 &= (\bar{A}\bar{B} + AB)C_{in} + (\bar{A}B + A\bar{B})\bar{C}_{in} \\
 &= (A \oplus B)C_{in} + (A \oplus B)\bar{C}_{in} \\
 &= (A \oplus B) \cdot C_{in} + (A \oplus B) \cdot \bar{C}_{in} \quad \{x = A \oplus B\} \\
 S &= x \oplus C_{in} = \boxed{A \oplus B \oplus C_{in}}
 \end{aligned}$$



K-Map for C

Cin \ AB				
	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$C = AB + AC_{in} + BC_{in}$$

$$\boxed{C = AB + C_{in}(A \oplus B)}$$

$$C = AB + \bar{A}BC_{in} + A\bar{B}C_{in}$$

$$= AB + C_{in}(\bar{A}B + A\bar{B})$$

$$= AB + C_{in}(A \oplus B)$$

$$+ \bar{A}\bar{B}\bar{C}_{in}$$

$$+ \bar{A}\bar{B}C_{in}$$

$$+ (\bar{A} \oplus \bar{B})\bar{C}_{in}$$

$$+ (A \oplus B)$$

$$2^n = 2^3 = 8$$

Designing of a Full Adder

A	B	C _{in}	S	C
0	0	0	0	0
1	0	0	0	0
1	0	1	0	1
0	1	0	1	0
1	1	0	0	1
0	1	1	1	1
1	1	1	1	0

K-Map for S

AB \ C _{in}	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$S = A \oplus B \oplus C_{in}$$

K-Map for C

AB \ C _{in}	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$C = AB + AC_{in} + BC_{in}$$

$$C = AB + C_{in}(A \oplus B)$$

$$C = AB + \bar{A}BC_{in} + A\bar{B}C_{in}$$

$$= AB + C_{in}(\bar{A}B + A\bar{B})$$

$$C = AB + C_{in}(A \oplus B)$$

